

3D Power Spectrum & Weak Lensing Theory

A Comprehensive Wiki with Spherical Fourier-Bessel Framework

Tirocinio MPMSSIA Internship Project

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Contents

0.1 Introduction

This wiki is a comprehensive, rigorously written exposition of the theory and methods underlying 3D power spectrum computations in cosmology, with emphasis on the **Spherical Fourier-Bessel (SFB) framework** for weak lensing and large-scale structure.

0.1.1 Motivation and Scope

Modern cosmological surveys measure the distribution of matter and galaxies across large volumes of the universe, as a function of both angular position on the sky and distance (through redshift). Traditional [Limber approximation](#) approaches reduce these 3D measurements to 2D angular power spectra, sacrificing information about the full 3D structure.

This wiki develops the complete mathematical and computational framework for:

- Computing **full 3D power spectra** without the Limber flat-sky approximation
- Decomposing 3D fields on the *Spherical Fourier-Bessel* basis, which naturally respects spherical geometry
- Connecting 3D power spectra to observable **weak lensing convergence and shear**
- Implementing efficient numerical integration schemes for cosmological observables
- Validating results against Limber benchmarks and physical consistency checks

0.1.2 Who Should Read This Wiki?

Cosmology students seeking a rigorous exposition of power spectrum theory beyond textbooks

Researchers working on large-scale structure or weak lensing with 3D information

Developers implementing numerical codes for cosmological computations

Internship participants in this project, needing a single reference for all theory

0.1.3 Organization and How to Use This Wiki

The wiki is organized into **seven chapters**:

1. **?? (??)**: Cosmological background evolution, distances, and expansion history
2. **?? (??)**: Power spectrum foundations in Cartesian Fourier space, statistical ensembles, and the matter fluctuation field
3. **?? (??)**: Introduction to spherical harmonics, spherical Bessel functions, and the *Spherical Fourier-Bessel* basis expansion

4. ?? (??): 3D power spectrum decomposed in *Spherical Fourier-Bessel* coordinates; relates to Limber and angular power spectra
5. ?? (??): Weak gravitational lensing geometry, convergence/shear fields, and their 3D power spectra
6. ?? (??): Observable angular power spectra from 3D fields; redshift binning; cross-spectra between tracers
7. ?? (??): Numerical integration strategies, quadrature rules, and performance considerations

0.1.4 How to Navigate

- ****Start here****: Read [Introduction](#) (this section), then ??
- ****Quick reference****: See ?? for common equations and formulas
- ****Concept index****: ?? lists all major concepts with page references
- ****Cross-links****: All sections are linked via clickable hyperlinks (blue text) and `\autoref` references like “[section 0.1](#)”
- ****PDF search****: Use your PDF viewer’s search to find specific terms (all concepts are indexed)

0.1.5 Prerequisites

Readers should be familiar with:

- Basic cosmology: Friedmann equations, scale factor, comoving coordinates
- Fourier analysis: plane waves, Fourier transforms, convolutions
- Spherical harmonics and orthonormal function spaces
- Perturbation theory and correlation functions
- Linear algebra and tensor notation

For those needing a refresher, references to standard textbooks appear throughout.

0.1.6 Notation and Conventions

Mathematical notation follows standard cosmology conventions (see ?? for complete reference):

- k denotes wavenumber in the Fourier domain

- ℓ is the angular multipole order (related to angular scales)
- z or a denotes redshift or scale factor
- χ is comoving distance
- $P(k)$ is the 3D isotropic power spectrum
- C_ℓ is the angular power spectrum
- All integrals are in natural units where $\hbar = c = 1$ unless otherwise stated

0.1.7 Structure of Each Section

Each topic is organized as follows:

1. **Conceptual introduction:** Physical meaning and intuition
2. **Mathematical definitions:** Rigorous equations with all terms explained
3. **Examples:** Concrete applications or numerical illustrations
4. **Links to related concepts:** Cross-references to prerequisites and dependent topics
5. **Remarks:** Physical interpretation, limitations, and extensions

0.1.8 On Maintaining This Wiki

This wiki is designed to be maintained and extended daily. See `README_WIKI.md` for detailed maintenance instructions:

- Each concept lives in a separate `.tex` file
- All cross-references use `\label` and `\autoref`
- New sections are added to appropriate folders without modifying other files
- The master document (`wiki_main.tex`) includes all sections in a fixed order

To add a new concept:

1. Create `SECTION/my_concept.tex`
2. Write with structure: `\section{...}\label{sec:...}`
3. Include in `wiki_main.tex`
4. Add entry to `index_concepts.tex`
5. Link from related concepts using `\autoref{}`

0.1.9 Key References

This wiki draws on:

- BLAST survey paper: [\[BLAST2024\]](#)
- Standard cosmology texts: Dodelson & Schmidt, Peebles, Liddle & Lyth
- Spherical Fourier-Bessel methods: [\[SFB_methods\]](#)
- Weak lensing theory: [\[WeakLensingReview\]](#)

All citations appear in the References section at the end.

0.1.10 Questions and Feedback

This wiki is a living document. Issues, clarifications, and additional examples are welcomed. Please refer to the repository documentation for contribution guidelines.

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Ready to dive in? Start with [??](#), then proceed to [??](#).

0.2 Notation and Conventions

This section establishes all mathematical notation, constants, and conventions used throughout the wiki. Refer back here whenever unfamiliar symbols appear.

0.2.1 Cosmological Parameters and Symbols

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Symbol	Meaning	Notes
H_0	Hubble constant	Present-day expansion rate; $\approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$
$H(z)$	Hubble parameter at redshift z	Expansion rate; related to scale factor
$E(z) \equiv H(z)/H_0$	Normalized Hubble parameter	Dimensionless; $E(0) = 1$
a	Scale factor	$a_0 = 1$ today; $a = 1/(1+z)$
z	Redshift	Measure of cosmic distance/age
Ω_m	Matter density parameter	$\Omega_m = \rho_m/\rho_{\text{crit}}$ today
Ω_Λ	Dark energy density parameter	$\Omega_\Lambda = \rho_\Lambda/\rho_{\text{crit}}$
Ω_b	Baryon density parameter	Fraction of matter that is baryons
$\bar{\rho}_m$	Mean matter density	$\rho_m(z)$ averaged over all space
$\chi(z)$	Comoving distance	Distance to redshift z ; in comoving Mpc
$D_+(z)$	Linear growth factor	Amplitude of linear perturbations
$D_A(z)$	Angular diameter distance	Relates angular size to physical size
$P(k)$	3D isotropic power spectrum	Cartesian Fourier; only depends on $ \mathbf{k} = k$
$C_\ell(k)$	3D power spectrum in <i>Spherical Fourier-Bessel</i> basis	Power spectrum in spherical coordinates
C_ℓ	Angular power spectrum	Projected 2D power spectrum on sky
$\xi(r)$	Correlation function	Fourier transform of $P(k)$; depends only on $r = \mathbf{r} $

0.2.2 Spatial and Spectral Coordinates

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Symbol	Type	Interpretation
$\mathbf{x}, \mathbf{r}, \mathbf{k}$	3D Cartesian vectors	Physical/comoving position; Fourier wavenumber
(x, y, z)	Cartesian components	In physical/comoving frame
(r, θ, ϕ)	Spherical coordinates	r is radial; $\theta \in [0, \pi]$ polar; $\phi \in [0, 2\pi)$ azimuthal
$k = \mathbf{k} $	Wavenumber magnitude	Related to scale by $\lambda = 2\pi/k$
ℓ	Angular multipole	Spherical harmonic order; $\ell = 0$ monopole, $\ell = 1$ dipole, etc.
m	Azimuthal number	$-\ell \leq m \leq \ell$
Ω	Solid angle	$\Omega = 4\pi$ steradians for full sky
$d\Omega = \sin\theta d\theta d\phi$	Solid angle element	On unit sphere

0.2.3 Basis Functions and Decompositions

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Symbol	Meaning	Notes
$Y_{\ell m}(\theta, \phi)$	Spherical harmonic	Orthonormal on unit sphere; $\int d\Omega Y_{\ell m} ^2 = 1$
${}_s Y_{\ell m}$	Spin- s spherical harmonic	For tensor fields; spin-weight describes rotational behavior
$j_\ell(x)$	Spherical Bessel function	Regular solution to radial Bessel equation
$n_\ell(x)$	Spherical Neumann function	Irregular (singular) solution; usually not used for regular field
$j'_\ell(x)$	Derivative of j_ℓ	$\frac{d}{dx} j_\ell(x)$
$x_{\ell, n}$	n -th zero of $j_\ell(x)$	Roots of j_ℓ ; used for SFB radial grid
$j_\ell(kr)Y_{\ell m}(\theta, \phi)$	SFB basis function	Complete orthonormal basis in $\mathbb{R}^3$

0.2.4 Field Decompositions

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Symbol	Expansion	Context
$\tilde{f}(\mathbf{k})$	Cartesian Fourier coefficient	$f(\mathbf{x}) = \frac{1}{(2\pi)^3} \int d^3\mathbf{k} \tilde{f}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}}$
$f_{\ell m}(k)$	SFB coefficient	Coefficient in expansion on $j_\ell(kr)Y_{\ell m}$ basis
$\delta_{\ell m}(k)$	Matter perturbation SFB coeff	For matter overdensity $\delta(\mathbf{r})$
$\kappa_{\ell m}$	Convergence harmonic coeff	For lensing convergence field
$\gamma_{\ell m}^E, \gamma_{\ell m}^B$	E/B-mode shear coefficients	Curl-free vs curl-part decomposition of shear

0.2.5 Weak Lensing Fields

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Symbol	Meaning	Physical Interpretation
$\varphi(\mathbf{r})$	Lensing potential	Integrated effect of gravitational potential on light rays
$\kappa(\mathbf{r})$	Convergence (magnification)	Isotropic magnification/demagnification due to lensing
$\gamma(\mathbf{r})$	Shear (tidal distortion)	Anisotropic tidal stretching; complex field $\gamma = \gamma_1 + i\gamma_2$
γ_1, γ_2	Cartesian shear components	Components in tangent-plane coordinates
$W^\kappa(z)$	Convergence kernel	Lensing efficiency for convergence
$W^\gamma(z)$	Shear kernel	Lensing efficiency for shear
$P_\kappa(k)$	Convergence power spectrum	Power spectrum of lensing convergence
$C_\ell^{\kappa\kappa}$	Convergence angular power spectrum	Angular power spectrum of convergence
$C_\ell^{\kappa\gamma}$	Convergence-shear cross spectrum	Cross-correlation between convergence and shear

0.2.6 Integration and Statistics

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Symbol	Meaning	Usage
$\langle \cdot \rangle$	Ensemble average	$\langle f(\mathbf{x})f(\mathbf{x}') \rangle$ over many universe realizations
$\delta(\mathbf{k})$	Dirac delta function	$\int d^3\mathbf{k} \delta^3(\mathbf{k}) f(\mathbf{k}) = f(\mathbf{0})$
$\delta(\mathbf{k} - \mathbf{k}')$	3D Dirac delta	Enforces $\mathbf{k} = \mathbf{k}'$
δ_{ij}	Kronecker delta	$= 1$ if $i = j$, else 0
$\mathcal{P}(z_1, z_2)$	Integral over all scales	$\mathcal{P}(z_1, z_2) = \int_0^\infty dk k^2 P(k, z_1, z_2)$
$\int_0^\infty dk$	Integral over wavenumber	From $k = 0$ (largest scales) to $k \rightarrow \infty$ (smallest scales)
$\int d\Omega$	Integral over solid angle	$\int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta = 4\pi$

0.2.7 Operators

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Operator	Definition	Action
$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$	3D Laplacian	Acts on functions of (x, y, z)
$\nabla_\perp^2$	Laplacian on sphere	$\nabla_\perp^2 f _{\text{sphere}} = -\ell(\ell+1)f_{\ell m}/r^2$ for $f = f_{\ell m} r^2 Y_{\ell m}$
$\mathcal{F}$	Fourier transform	$\mathcal{F}[f] = \tilde{f}$ (from $\mathbf{x}$ to $\mathbf{k}$ space)
$\mathcal{F}^{-1}$	Inverse Fourier transform	$\mathcal{F}^{-1}[\tilde{f}] = f$ (from $\mathbf{k}$ to $\mathbf{x}$ space)
$\otimes$	Convolution	$(f \otimes g)(r) = \int d^3r' f(r')g(\mathbf{r} - \mathbf{r}')$

0.2.8 Physical Constants

We use natural units where $\hbar = c = 1$ throughout, unless otherwise specified. When needed:

- Speed of light: $c = 2.998 \times 10^8 \text{ m/s}$
- Gravitational constant: $G = 6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$
- Reduced Planck constant: $\hbar = 1.055 \times 10^{-34} \text{ J} \cdot \text{s}$
- Planck mass: $M_P = \sqrt{\hbar c/G} \approx 1.22 \times 10^{19} \text{ GeV}$

0.2.9 Units and Scales

- **Wavenumbers:** Expressed in Mpc^{-1} (megaparsec inverse) or $h \text{ Mpc}^{-1}$ where $h = H_0/(100 \text{ km/s/Mpc})$
- **Distances:** Comoving Mpc unless specified as physical (proper) distance

- **Power spectra:** $P(k)$ has units $(\text{Mpc})^3$ (volume in Fourier space)
- **Angular power spectra:** C_ℓ is dimensionless
- **Redshift:** Pure number; $z \approx 0$ at present epoch

0.2.10 Common Abbreviations

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Abbreviation Meaning

<i>Statistical Isotropy and Homogeneity</i>	Statistical Isotropy and Homogeneity
<i>Limber approximation</i>	Limber approximation (flat-sky, small-angle)
<i>Spherical Fourier-Bessel</i>	Spherical Fourier-Bessel framework
LSS	Large-scale structure
CMB	Cosmic Microwave Background
WL	Weak lensing
$N(z)$	Galaxy redshift distribution
LSST	Large Synoptic Survey Telescope
BLAST	Name of reference survey
BAO	Baryon Acoustic Oscillations
RSD	Redshift-space distortions

0.2.11 A Quick Sanity Check

If you see an equation like:

$$C_\ell(k_1, k_2) = \int_0^\infty \frac{d\chi}{\chi^2} W(\chi) j_\ell(k_1 \chi) j_\ell(k_2 \chi) \quad (1)$$

You should interpret it as: “The 3D power spectrum coefficient C_ℓ (depending on two wavenumbers k_1, k_2) is computed by integrating over comoving distance χ , with a weight function $W(\chi)$ and spherical Bessel functions j_ℓ .”

—
Next: Proceed to ?? for recommendations on how to read this wiki.

0.3 Reading Guide

Different readers have different backgrounds and goals. This guide suggests reading paths through the wiki depending on your needs.

0.3.1 Path 1: Complete Beginner in Cosmology

If you're new to cosmology and power spectra, follow this order:

1. ?? (??): Learn the expanding universe, distances, and growth of structure
2. ??, Section “Cartesian Fourier”: Power spectrum basics in flat space
3. ??, Section “Statistical Framework”: Ensemble statistics
4. ??, Section “Matter Perturbations”: What we're measuring
5. ?? (??): Spherical coordinates and basis functions
6. ?? (??): 3D power spectrum in spherical coordinates

Estimated reading time: 2–3 weeks (assuming 30–60 min/day)

0.3.2 Path 2: Experienced Cosmologist, New to SFB

If you know power spectra but want to learn the SFB framework:

1. Skim ??: Refresh on distances (χ , D_A , growth D_+)
2. Skim ??: You know this material already
3. ??: Full reading—this is the new material
4. ??: Connect SFB to observables
5. ??, Sections “Angular Power Spectrum” and “SFB Kernels”: See how SFB predicts observations

Estimated reading time: 3–5 days

0.3.3 Path 3: Weak Lensing Focus

If you're interested in weak lensing specifically:

1. ??, Section “Comoving Distance”: Need distances for lensing kernels
2. ??, Sections on “Spherical Harmonics”: Required for lensing fields
3. ??: Complete reading

4. ??, Sections on “Redshift Binning” and “Cross-Spectra”: Practical observables
5. ??: Numerical integration for lensing kernels

Estimated reading time: 4–5 days

0.3.4 Path 4: Numerical Implementation

If you’re implementing code:

1. ??, Sections on “Bessel Functions”: You’ll code this
2. ??: Understand what you’re computing
3. ??, Section “SFB Kernels”: The integrals you need to implement
4. ??: All of it—critical for your code
5. Return to ?? for kernel definitions (if including lensing)

Estimated reading time: 1–2 weeks (+ coding time)

0.3.5 Quick Reference for Specific Questions

- “What is the Limber approximation?” → ??, find “Limber”
- “How do redshift bins affect power spectra?” → ??, Section “Redshift Binning”
- “What’s the difference between convergence and shear?” → ??, Section “Convergence and Shear”
- “How do I compute C_ℓ from $P(k)$?” → ??, find projection integral
- “Which quadrature rule should I use?” → ??, Section “Quadrature Strategies”
- “What are E-modes and B-modes?” → ??, Section “Spin-Weight Harmonics”

Use the ****Concept Index**** (??) to find all relevant sections for a topic.

0.3.6 Study Tips

1. **Take notes:** Write equations in your own words; copy derivations by hand
2. **Check dimensions:** Every equation should have consistent units—this catches errors
3. **Follow cross-links:** When you see a reference like ??, click to understand context
4. **Work through examples:** If an example is given, compute it yourself before reading the answer

5. **Question assumptions:** When a result is derived, ask “why is this assumption valid?”
6. **Connect to code:** If you’re implementing, look at your code as you read—does it match the math?

0.3.7 Common Misconceptions (Clarified)

“SFB is just a change of coordinates” Partially true, but SFB naturally incorporates spherical geometry and boundary conditions in a way Cartesian Fourier does not for spherical domains.

“3D power spectra are always larger than angular power spectra” No—they contain different information. A 3D power spectrum can have more or less power in some modes depending on how information is distributed in redshift.

“Limber approximation is always accurate for large ℓ ” Yes for $\ell \gg 1$, but the error depends on redshift distribution and wavenumber range. See ?? for limits.

“Weak lensing is only important for small angles” Weak lensing is a full-sky effect. Small-angle approximations (Limber) are separately motivated and orthogonal to the lensing approximation.

“B-modes always indicate systematics” Pure gravitational lensing produces only E-modes, but other physical effects (primordial gravitational waves, intrinsic alignments in some models) can produce B-modes. It’s an observational diagnostic.

0.3.8 Updating Your Knowledge as the Wiki Grows

This wiki is maintained continuously. When new sections are added:

1. Check the Table of Contents for new chapters or sections
2. Look for references to new material in existing sections (they’ll be linked)
3. Use the ?? to find related concepts

Next: See ?? for a handy reference of key equations and formulas, then jump into the main content.

0.4 Quick Reference: Key Equations

This section collects the most frequently used equations and formulas for quick lookup.

0.4.1 Cosmological Background

$$E(z) = \frac{H(z)}{H_0} = [\Omega_m(1+z)^3 + \Omega_\Lambda]^{1/2} \quad (2)$$

$$\chi(z) = \int_0^z \frac{dz'}{H(z')/H_0} = \int_0^z \frac{dz'}{E(z')} \quad (3)$$

$$D_A(z) = \frac{\chi(z)}{1+z} \quad (4)$$

0.4.2 Power Spectrum: Cartesian Fourier

$$\langle \tilde{\delta}(\mathbf{k}) \tilde{\delta}^*(\mathbf{k}') \rangle = (2\pi)^3 P(k) \delta^3(\mathbf{k} - \mathbf{k}') \quad (5)$$

$$\xi(r) = \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k) j_0(kr) \quad (6)$$

0.4.3 Spherical Decomposition

$$f(\mathbf{r}) = \sqrt{\frac{2}{\pi}} \int_0^\infty dk \sum_{\ell m} f_{\ell m}(k) k j_\ell(kr) Y_{\ell m}(\theta, \phi) \quad (7)$$

$$f_{\ell m}(k) = \sqrt{\frac{2}{\pi}} \int_0^\infty dr r^2 \int d\Omega f(\mathbf{r}) k j_\ell(kr) Y_{\ell m}^*(\theta, \phi) \quad (8)$$

$$\int_0^\infty j_\ell(kr) j_\ell(k'r) r^2 dr = \frac{\pi}{2k^2} \delta(k - k') \quad (9)$$

0.4.4 3D Power Spectrum in SFB

$$\langle f_{\ell m}(k) f_{\ell' m'}^*(k') \rangle = C_\ell(k) \delta(k - k') \delta_{\ell\ell'} \delta_{mm'} \quad (10)$$

Under *Statistical Isotropy and Homogeneity*:

$$C_\ell(k) = P(k) \quad (\text{independent of } \ell, m) \quad (11)$$

0.4.5 Angular Power Spectrum (Limber)

$$C_\ell^{AB} = \int_0^\infty \frac{d\chi}{\chi^2} W_A(\chi) W_B(\chi) P(k = (\ell + 1/2)/\chi, \chi) \quad (12)$$

where $W_A(\chi), W_B(\chi)$ are lensing kernels for tracers A, B.

0.4.6 Weak Lensing Fields

Convergence from potential:

$$\kappa_{\ell m} = \frac{\ell(\ell+1)}{2} \varphi_{\ell m} \quad (13)$$

Convergence power spectrum:

$$P_\kappa(k) = \left(\int dz \frac{dn}{dz} W^\kappa(z) \right)^2 P(k) \quad (14)$$

Convergence kernel:

$$W^\kappa(z) = \frac{3}{2} H_0^2 \Omega_m (1+z) D_+(z) \int_z^\infty dz' \frac{dn}{dz'} \frac{\chi(z') - \chi(z)}{\chi(z')} \quad (15)$$

0.4.7 Numerical Integration Formulas

Gauss-Legendre quadrature (adaptive):

$$\int_a^b f(x) dx \approx \sum_{i=1}^n w_i f(x_i) \quad (16)$$

Where x_i and w_i are quadrature nodes and weights, n is order.

Spherical Bessel recurrence:

$$j_{\ell-1}(x) + j_{\ell+1}(x) = \frac{2\ell+1}{x} j_\ell(x) \quad (17)$$

0.4.8 Commonly Used Numbers

- $H_0 \approx 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Planck 2018)
- $\Omega_m \approx 0.315$ (matter density today)
- $\Omega_\Lambda \approx 0.685$ (dark energy today)
- $\Omega_b \approx 0.049$ (baryon density today)
- $D_+ \approx a$ at late times (scale factor approximation)
- $z \sim 0.5\text{--}2.0$ typical for galaxy surveys
- $k \sim 0.01\text{--}1 \text{ h Mpc}^{-1}$ typical for cosmological scales

0.4.9 Quick Conversions

- Wavenumber to scale: $\lambda = 2\pi/k$
- Angular scale to multipole: $\ell \approx \pi/\theta$ (for small θ in radians)
- Redshift to scale factor: $a = 1/(1+z)$

- $h = H_0/(100 \text{ km s}^{-1} \text{ Mpc}^{-1})$ (reduced Hubble)

Tip: Bookmark this page for reference while reading the main content. Use **Ctrl+F** to search for specific equations.

0.5 Concept Index

This index lists all major concepts discussed in the wiki, organized alphabetically with cross-references and page numbers. Use this to find topics quickly or discover related concepts.

0.5.1 A

Angular Diameter Distance ??; See ??, Distance section

Angular Multipole (ℓ) ??; See ??, Spherical Harmonics section

Angular Power Spectrum (C_ℓ) ??; See ??, Angular Power section

Anisotropic Clustering See Redshift-Space Distortions

Azimuthal Mode (m) ??

0.5.2 B

Baryon Acoustic Oscillations See ??, Matter Perturbations section

Bessel Functions Spherical: ??; See ??, Bessel Functions section

B-modes See ??, Spin-Weight Harmonics section

Background Evolution ??; See ??

0.5.3 C

Cartesian Fourier See ??, Cartesian Fourier section

Chebyshev Interpolation See ??, Integration Schemes section

Comoving Distance ??; See ??, Distance section

Convergence ??; See ??, Convergence and Shear section

Convergence Kernel ??; See ?? and ??

Convergence Power Spectrum ??; See ??, Power Spectrum section

Correlation Function ??; See ??, Statistical Framework section

Cross-spectrum See ??, Cross-Spectra section

0.5.4 D

Density Perturbations See ??, Matter Perturbations section

Dirac Delta Function ??

Distance Measures See ??, all sections

0.5.5 E

E-modes See ??, Spin-Weight Harmonics section

Ensemble Average ??

Expansion History See ??, Background Evolution section

Expansion Parameters $\Omega_{\text{m}}, \Omega_{\Lambda}, \Omega_{\text{b}}$: ??

0.5.6 F

Fourier Analysis See ??, Cartesian Fourier section

Fourier Transform ??

Friedmann Equations See ??, Cosmological Model section

Full Sky Spherical coordinates essential; See ??

0.5.7 G

Galaxy Clustering See ??, Matter Perturbations section

Gauss-Legendre Quadrature ??; See ??, Quadrature Strategies section

Gravitational Lensing See ??

Growth Factor D_+ : ??; See ??, Background Evolution section

0.5.8 H

Hubble Parameter $H(z)$: ??; See ??, Background Evolution section

Hubble Constant H_0 : ??

0.5.9 I

Intrinsic Alignments See ??, Remarks section

Isotropic Power Spectrum ??; See ??, Statistical Framework section

Isotropy and Homogeneity See ??, Statistical Framework section

0.5.10 K

Kernels, Integration See ??, SFB Kernels section

Kernels, Lensing Convergence: ??; Shear: See ??

Kronecker Delta ??

0.5.11 L**Laplacian on Sphere** ??**Large-Scale Structure** See ??**Lensing Efficiency** See **Kernels****Lensing Potential** See ??, **Lensing Geometry** section**Limber Approximation** ??; See ?? and ??**Linear Growth** See **Growth Factor****0.5.12 M****Magnification** See **Convergence****Matter Density Parameter** Ω_m : ??**Matter Power Spectrum** See **Power Spectrum****Multipole Expansion** See **Spherical Harmonics** and **Angular Multipole****0.5.13 N****Neumann Function** ??**Numerical Integration** See ??**0.5.14 O****Observable Clustering** See ??**Orthogonal Basis** ??; See ??, **Orthonormality** section**0.5.15 P****Planck Satellite** See ??, **references****Power Spectrum** See **Power Spectrum (3D)**, **Angular Power Spectrum**, or specific variants**Power Spectrum, 3D** ??; See ??**Power Spectrum, 3D in SFB** ??; See ??**Projected Power** See **Angular Power Spectrum**

0.5.16 Q

Quadrature Rules See ??, Quadrature Strategies section

Quasi-Linear Scales See ??, Matter Perturbations section

0.5.17 R

Radial Grid Points Zeros of Bessel: ??; See ??, SFB Expansion section

Redshift z : ??

Redshift Binning See ??, Redshift Binning section

Redshift Distribution $n(z)$; See ??, Redshift Binning section

Redshift-Space Distortions See ??, Matter Perturbations section

0.5.18 S

Scale Factor a : ??

Shear ??; See ??, Convergence and Shear section

Small-Angle Approximation See Limber Approximation

Solid Angle ??

Spherical Bessel Functions See Bessel Functions

Spherical Fourier-Bessel See ?? and ??

Spherical Harmonics See ??, Spherical Harmonics section

Spin Weight See ??, Spin-Weight Harmonics section

Statistical Isotropy and Homogeneity See ??, Statistical Framework section

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Tensor Fields Shear is spin-2; See ??

Transfer Function See ??, Matter Perturbations section

Tidal Distortion See Shear

0.5.20 U

Universe, Expansion of See ??

0.5.21 W

Wavenumber k : ??

Weak Lensing See ??

Wiki Maintenance See README_WIKI.md

0.5.22 Z

Zero of Bessel Function Radial grid points; See ??, SFB Expansion section

How to use this index:

1. Look up a concept alphabetically
2. Click on any page reference (blue text) to jump to that section
3. Related concepts are listed under “See also” entries

Not finding what you need? Use **Ctrl+F** to search the PDF for keywords, or check the Table of Contents for chapter overviews.

Chapter 1

Cosmological Background

1.0.1 Cosmological Model and Parameters

This subsection introduces the standard cosmological model and key parameters that describe the expanding universe.

The universe is assumed to be homogeneous and isotropic on large scales (above ~ 100 Mpc). Under the assumption of spatial flatness (consistent with observations), the metric is:

$$ds^2 = -dt^2 + a(t)^2(dx^2 + dy^2 + dz^2) \quad (1.1)$$

where $a(t)$ is the scale factor and (x, y, z) are comoving coordinates.

1.0.1.1 Friedmann Equations

The evolution of $a(t)$ is governed by Einstein's field equations in the form of the Friedmann equations:

$$H(a)^2 \equiv \left(\frac{1}{a} \frac{da}{dt}\right)^2 = \frac{8\pi G}{3} \rho_{\text{tot}}(a) \quad (1.2)$$

$$\frac{1}{a} \frac{d^2 a}{dt^2} = -\frac{4\pi G}{3} (\rho + 3p) \quad (1.3)$$

where $H(a)$ is the Hubble parameter, ρ is density, and p is pressure.

1.0.1.2 Density Parameters

For a flat universe, we define dimensionless density parameters:

$$\Omega_m = \frac{\rho_m}{\rho_c}, \quad \Omega_\Lambda = \frac{\rho_\Lambda}{\rho_c}, \quad \Omega_b = \frac{\rho_b}{\rho_c} \quad (1.4)$$

where $\rho_c = 3H_0^2/(8\pi G)$ is the critical density today.

Flatness requires $\Omega_m + \Omega_\Lambda + \Omega_k = 1$ (where Ω_k is curvature density).

For the current epoch ($z = 0$):

- $\Omega_{\text{m}} \approx 0.315$ (mostly dark matter)
- $\Omega_{\Lambda} \approx 0.685$ (dark energy, likely cosmological constant)
- $\Omega_{\text{b}} \approx 0.049$ (baryons only)

1.0.1.3 Normalized Hubble Parameter

It is convenient to work with the normalized Hubble parameter:

$$E(a) = \frac{H(a)}{H_0} = \sqrt{\Omega_{\text{m}}a^{-3} + \Omega_{\Lambda} + \Omega_{\text{k}}a^{-2}} \quad (1.5)$$

or in terms of redshift $z = 1/a - 1$:

$$E(z) = \sqrt{\Omega_{\text{m}}(1+z)^3 + \Omega_{\Lambda}} \quad (1.6)$$

(assuming flatness, $\Omega_{\text{k}} = 0$).

Remark 1.1. The Hubble constant $H_0 \approx 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Planck 2018). This is one of the most precisely measured quantities in cosmology, though recent measurements show tension with local measurements.

1.0.1.4 Redshift Relation

Redshift is related to the scale factor by:

$$1 + z = \frac{1}{a} \quad (1.7)$$

At present ($z = 0$), $a = 1$. At earlier times, $a < 1$ and $z > 0$. This provides a direct observational window into the universe's expansion history.

See ?? for how $a(t)$ evolves with time.

1.0.2 Background Evolution

This subsection describes how the scale factor and Hubble parameter evolve with cosmic time, and introduces the concept of lookback time.

1.0.2.1 Scale Factor Evolution

For a flat CDM universe with equation of state w_i for each component, the scale factor satisfies:

$$\frac{da}{dt} = H_0 a \sqrt{\Omega_m a^{-3} + \Omega_\Lambda} \quad (1.8)$$

This differential equation has no closed-form solution in general. However, several limiting cases are instructive:

Matter-dominated era ($a \ll a_{\text{eq}}$, early universe):

$$a(t) \propto t^{2/3}, \quad H(t) = \frac{2}{3t} \quad (1.9)$$

Dark energy-dominated era ($a \gg a_{\text{eq}}$, late universe):

$$a(t) \propto e^{H_0 \sqrt{\Omega_\Lambda} t}, \quad H \rightarrow \sqrt{\Omega_\Lambda} H_0 = \text{const} \quad (1.10)$$

1.0.2.2 Lookback Time

The lookback time to redshift z (the time difference between now and when the universe was at redshift z) is:

$$t_{\text{lookback}}(z) = \int_0^z \frac{dz'}{(1+z')H(z')} \quad (1.11)$$

This is useful for understanding which epochs we observe at which redshifts.

Remark 1.2. At $z \approx 0.5$ (accessible to many surveys), we look back ≈ 5 Gyr. At $z \sim 2$ (high-redshift surveys), we look back ≈ 10 Gyr, catching structure formation at earlier epochs.

1.0.2.3 Age of the Universe

The age of the universe today is:

$$t_0 = \int_0^\infty \frac{dz}{(1+z)H(z)} \approx 13.8 \text{ Gyr} \quad (1.12)$$

See ?? for the definition of $E(z)$ that enters $H(z) = H_0 E(z)$.

1.0.3 Distance Measures

This subsection defines the various distance measures used in cosmology. These are essential for converting observations (angles on the sky, apparent magnitudes) into physical quantities.

1.0.3.1 Comoving Distance

The comoving distance $\chi(z)$ to an object at redshift z is the distance accounting for the expansion of the universe. It is defined as:

$$\chi(z) = \int_0^z \frac{c \, dz'}{H(z')} = \int_0^z \frac{dz'}{E(z')} \quad (1.13)$$

(In natural units where $c = H_0 = 1$.)

At large redshift, $\chi(z)$ approaches $\chi_\infty \approx 2c/H_0 \approx 14,000$ Mpc (the horizon distance).

Remark 1.3. Comoving distance is the natural distance scale for cosmological surveys because it removes the effect of cosmic expansion. Two galaxies at the same redshift but different angular positions on the sky have comoving separation that is independent of when we observe them.

1.0.3.2 Angular Diameter Distance

The angular diameter distance $D_A(z)$ relates the angular size θ (in radians) to the physical size s of an object:

$$\theta = \frac{s}{D_A(z)} \quad (1.14)$$

For a flat universe:

$$D_A(z) = \frac{\chi(z)}{1+z} \quad (1.15)$$

The factor $(1+z)$ accounts for the redshift dimming: sources at higher redshift appear smaller and dimmer.

1.0.3.3 Luminosity Distance

The luminosity distance $D_L(z)$ relates the absolute luminosity L of a source to the observed flux F :

$$F = \frac{L}{4\pi D_L(z)^2} \quad (1.16)$$

For a flat universe:

$$D_L(z) = (1+z)\chi(z) \quad (1.17)$$

Note that $D_L(z) = (1+z)^2 D_A(z)$. This duality is a consequence of the reciprocity theorem in general relativity.

1.0.3.4 Comoving Volume Element

The comoving volume element per unit solid angle per unit redshift is:

$$\frac{dV_c}{d\Omega dz} = \frac{\chi(z)^2}{H(z)} \quad (1.18)$$

This is essential for computing the number density of galaxies and other volume statistics.

Example 1.1. For $\Omega_m = 0.3, \Omega_\Lambda = 0.7, H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$:

- $\chi(0.5) \approx 1547 \text{ Mpc}$
- $D_A(0.5) \approx 1031 \text{ Mpc}$
- $\left. \frac{dV_c}{d\Omega dz} \right|_{z=0.5} \approx 1.1 \times 10^9 \text{ Mpc}^3 \text{ deg}^{-2}$

1.0.3.5 Relating Observations to Physical Quantities

- **Angular scale on sky:** A small angle θ corresponds to physical scale $s = \theta D_A(z)$
- **Apparent magnitude:** Related to luminosity distance via $m - M = 5 \log_{10} D_L(z) - 5$
- **Galaxy counts:** Number of galaxies in a redshift bin depends on comoving volume via ??

These distance measures are fundamental inputs to weak lensing kernels (??) and observables computations (??).

See ?? for the definition of $E(z)$ and ?? for lookback time.

Chapter 2

Power Spectrum: Fundamentals

2.0.1 Cartesian Fourier Transform and Plane Waves

This subsection introduces the standard Cartesian Fourier decomposition of 3D fields, which serves as a reference point for all other decompositions.

2.0.1.1 Field expansion in plane waves

Any square-integrable scalar field $f(\mathbf{x})$ defined in 3D Euclidean space can be expanded as:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^3} \int d^3\mathbf{k} \tilde{f}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}} \quad (2.1)$$

The inverse transformation extracts the Fourier coefficients:

$$\tilde{f}(\mathbf{k}) = \int d^3\mathbf{x} f(\mathbf{x}) e^{-i\mathbf{k}\cdot\mathbf{x}} \quad (2.2)$$

where $\mathbf{k}$ is the wave vector, related to wavelength by $\lambda = 2\pi/|\mathbf{k}|$.

2.0.1.2 Isotropy: dependence on magnitude only

For an isotropic field (same statistics in all directions), the Fourier coefficients depend only on $k = |\mathbf{k}|$, not on direction:

$$\tilde{f}(\mathbf{k}) = f(k) \cdot Y(\hat{\mathbf{k}}) \quad (2.3)$$

where $\hat{\mathbf{k}} = \mathbf{k}/k$ is the direction and Y is an angular weight function (e.g., spherical harmonic).

Remark 2.1. This isotropy is fundamental in cosmology: the universe looks the same in all directions (at least on large scales), so we expect all statistical quantities to be isotropic.

See ?? for complete notation conventions and ?? for power spectrum definitions.

2.0.2 Statistical Framework: Ensemble Averages and Power Spectra

This subsection formalizes the statistical framework for power spectra using ensemble averaging over possible universe realizations.

2.0.2.1 Ensemble average and ergodicity

For a random field like cosmic density fluctuations, we define ensemble averages over all possible realizations of the universe (even though we observe only one). Under the cosmological principle (isotropy and homogeneity), ensemble statistics are equivalent to spatial averages for stationary fields:

$$\langle f(\mathbf{x})g(\mathbf{x}') \rangle = \text{Ensemble average} \quad (2.4)$$

For homogeneous fields, the two-point correlation function depends only on separation $\mathbf{r} = \mathbf{x} - \mathbf{x}'$:

$$\xi(r) = \langle f(\mathbf{x})f(\mathbf{x} + \mathbf{r}) \rangle - \langle f \rangle^2 \quad (2.5)$$

2.0.2.2 Definition of power spectrum

The power spectrum is the Fourier transform of the correlation function (Wiener-Khinchin theorem):

$$P(k) = \int d^3\mathbf{r} \xi(r) e^{i\mathbf{k}\cdot\mathbf{r}} \quad (2.6)$$

Equivalently, in terms of Fourier coefficients:

$$\langle \tilde{f}(\mathbf{k})\tilde{f}^*(\mathbf{k}') \rangle = (2\pi)^3 P(k) \delta^3(\mathbf{k} - \mathbf{k}') \quad (2.7)$$

For isotropic fields, $P(k)$ depends only on wavenumber magnitude $k = |\mathbf{k}|$.

Remark 2.2. The delta function enforces that different wavenumber modes are uncorrelated (by isotropy). The factor $(2\pi)^3$ is a normalization convention arising from Fourier conventions.

2.0.2.3 Dimensional analysis

In 3D Fourier space: - $P(k)$ has dimensions of $[\text{length}]^3$ (volume in Fourier space) - $P(k) dk$ represents the contribution to total variance from modes in the range $[k, k + dk]$

The total variance is:

$$\sigma^2 = \int_0^\infty \frac{dk}{2\pi^2} k^2 P(k) \quad (2.8)$$

See ?? for Cartesian Fourier conventions and ?? for spherical decompositions.

2.0.3 Matter Perturbations and Structure Growth

This subsection describes the matter power spectrum and how it evolves with cosmic time.

2.0.3.1 Linear perturbation theory

At early times, density perturbations $\delta(\mathbf{x}, z) = (\rho - \bar{\rho})/\bar{\rho}$ are small and grow linearly:

$$\delta(k, z) = D_+(z)\delta(k, z=0) \quad (2.9)$$

where $D_+(z)$ is the linear growth factor, normalized to 1 today.

The matter power spectrum evolves as:

$$P(k, z) = D_+^2(z) P(k, z=0) = D_+^2(z) P_0(k) \quad (2.10)$$

where $P_0(k)$ is a reference spectrum (usually at $z=0$ or early times).

2.0.3.2 Linear vs nonlinear scales

- ****Linear scales**** ($k < k_{\text{nl}}$): Perturbation theory accurate; $P(k) \propto D_+^2$ - ****Nonlinear scales**** ($k > k_{\text{nl}}$): Mode coupling; power enhanced; requires N -body simulations or fitting formulas

The nonlinear scale $k_{\text{nl}}(z)$ increases with redshift (less time for collapse):

$$k_{\text{nl}}(z) \approx 0.12 h \text{ Mpc}^{-1} \quad (z=0) \quad \rightarrow \quad 0.25 h \text{ Mpc}^{-1} \quad (z=2) \quad (2.11)$$

2.0.3.3 Galaxy bias

Galaxies do not trace the matter field 1-to-1. Linear galaxy bias relates galaxy and matter power spectra:

$$P_{\text{gal}}(k, z) = b^2(z) P_{\text{matter}}(k, z) \quad (2.12)$$

where $b(z)$ is the linear bias parameter ($b > 1$ for rare, massive objects; $b \approx 1$ for typical galaxies).

Example 2.1. CDM with $\Omega_{\text{m}} = 0.3, \Omega_{\Lambda} = 0.7$ gives: - At $z=0$: $D_+(0) \approx 1$ (normalized) - At $z=1$: $D_+(1) \approx 0.76$ (structure grows more slowly at high- z) - Typical galaxy bias: $b \approx 1.5$ – 2 at $z \sim 1$

See ?? for growth factor evolution and ?? for weak lensing kernels involving matter power.

2.0.4 Spherical Decomposition of Isotropic Fields

This section bridges Cartesian Fourier and spherical coordinates, explaining how an isotropic field decomposes in both bases.

2.0.4.1 Completeness of spherical harmonics

Any function on the unit sphere can be expanded in spherical harmonics:

$$f(\theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} f_{\ell m} Y_{\ell m}(\theta, \phi) \quad (2.13)$$

For isotropic fields, the power depends only on ℓ :

$$\langle f_{\ell m} f_{\ell' m'}^* \rangle = C_{\ell} \delta_{\ell \ell'} \delta_{m m'} \quad (2.14)$$

This ℓ -only dependence is the key to reducing 3D power to angular power spectra.

2.0.4.2 Spherical Fourier-Bessel (SFB) basis

For 3D fields in spherical coordinates (r, θ, ϕ) , the natural orthonormal basis combines radial and angular:

$$\text{Basis function: } j_{\ell}(kr) Y_{\ell m}(\theta, \phi) \quad (2.15)$$

where j_{ℓ} are spherical Bessel functions and k is the radial wavenumber.

Any field expands as:

$$f(\mathbf{r}) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} dk \sum_{\ell m} f_{\ell m}(k) k j_{\ell}(kr) Y_{\ell m}(\theta, \phi) \quad (2.16)$$

See ?? for complete treatment of Bessel functions and SFB orthonormality.

Chapter 3

Spherical Fourier-Bessel Framework

3.0.1 Spherical Bessel Functions

This section introduces spherical Bessel functions, which form the radial basis for 3D field decompositions in spherical coordinates.

3.0.1.1 Definition and properties

The spherical Bessel function of order ℓ is defined by:

$$j_\ell(x) = \sqrt{\frac{\pi}{2x}} J_{\ell+1/2}(x) \quad (3.1)$$

where J_ν is the Cylindrical Bessel function. Alternatively, a recurrence relation defines them:

$$j_\ell(x) = \frac{\ell}{x} j_{\ell-1}(x) - j'_{\ell-1}(x) \quad (3.2)$$

with initial conditions:

$$j_0(x) = \frac{\sin x}{x} \quad (3.3)$$

$$j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x} \quad (3.4)$$

3.0.1.2 Asymptotic behavior

- **Small argument** ($x \ll \ell$): $j_\ell(x) \approx \frac{x^\ell}{(2\ell+1)!!}$ - **Large argument** ($x \gg \ell$): $j_\ell(x) \approx \frac{\sin(x-\ell\pi/2)}{x}$

3.0.1.3 Zeros and roots

The n -th zero of $j_\ell(x)$ is denoted $x_{\ell,n}$. These are essential for defining SFB radial grids.

Example 3.1. For $\ell = 0$: $x_{0,1} = \pi, x_{0,2} = 2\pi, \dots$

For $\ell = 1$: $x_{1,1} \approx 4.49, x_{1,2} \approx 7.73, \dots$

See ?? for use in defining the SFB radial grid.

3.0.2 Spherical Fourier-Bessel Basis Functions

This section introduces the complete orthonormal basis of spherical Fourier-Bessel functions and their key properties.

3.0.2.1 The SFB basis set

The SFB basis functions are products of radial and angular components:

$$\psi_{\ell m}^{(n)}(\mathbf{r}) = A_{\ell n} j_{\ell}(k_{\ell n} r) Y_{\ell m}(\theta, \phi) \quad (3.5)$$

where: - j_{ℓ} is the spherical Bessel function of order ℓ - $Y_{\ell m}$ is the spherical harmonic - $k_{\ell n} = x_{\ell, n}/R$ with $x_{\ell, n}$ the n -th zero of j_{ℓ} - R is the domain radius (often taken to infinity) - $A_{\ell n}$ is a normalization constant

3.0.2.2 Domain and boundary conditions

For cosmological fields defined on $r \in [0, \infty)$: - Regularity at origin: $j_{\ell}(0) = 0$ for $\ell > 0$ - Decay at infinity: $j_{\ell}(kr) \rightarrow \cos(kr - \ell\pi/2)/r$

This makes Bessel functions natural for fields with no sources at infinity.

3.0.2.3 Advantages over plane waves

- **Spherical symmetry**: Respects the observer-centric geometry of surveys
- **Finite domains**: Natural handling of radial selection functions
- **Boundary conditions**: Automatic regularity at origin and decay at large r
- **Angular locality**: Spherical harmonics separate angular and radial information

See ?? for orthonormality relations and ?? for field expansions.

3.0.3 SFB Field Expansion and Inversion

This section shows how to decompose a 3D field in the SFB basis and recover the field from its coefficients.

3.0.3.1 Forward transformation: Field to coefficients

Any square-integrable field $f(\mathbf{r})$ in spherical coordinates can be decomposed as:

$$f_{\ell m}(k) = \sqrt{\frac{2}{\pi}} \int_0^\infty dr r^2 \int d\Omega f(\mathbf{r}) k j_\ell(kr) Y_{\ell m}^*(\theta, \phi) \quad (3.6)$$

This extracts the SFB coefficient $f_{\ell m}(k)$ for each multipole ℓ , azimuthal mode m , and radial wavenumber k .

3.0.3.2 Inverse transformation: Coefficients to field

The field is reconstructed from its coefficients by:

$$f(\mathbf{r}) = \sqrt{\frac{2}{\pi}} \int_0^\infty dk \sum_{\ell m} f_{\ell m}(k) k j_\ell(kr) Y_{\ell m}(\theta, \phi) \quad (3.7)$$

The factor $k dk$ in the measure accounts for the Jacobian of spherical Bessel function orthonormality.

3.0.3.3 Physical interpretation

- $f_{\ell m}(k)$ tells us the amplitude of the field at: - Radial scale $\sim 1/k$ (wavenumber k) - Angular scale corresponding to multipole ℓ - Azimuthal orientation m
- In cosmology: $f_{\ell m}(k)$ encodes how density fluctuations are distributed in redshift (via k) and angular multipole (via ℓ, m)

See ?? for why these expansions are complete and orthonormal.

3.0.4 Orthonormality and Completeness

This section establishes that the SFB basis is orthonormal and complete, justifying its use for decomposing any field.

3.0.4.1 Orthonormality of Bessel functions

The spherical Bessel functions satisfy:

$$\int_0^\infty r^2 j_\ell(kr) j_\ell(k'r) dr = \frac{\pi}{2k^2} \delta(k - k') \quad (3.8)$$

This relation is the analog of orthonormality for plane waves: $\int e^{ikx} e^{ik'x} dx \propto \delta(k - k')$.

3.0.4.2 Orthonormality of spherical harmonics

Standard spherical harmonics satisfy:

$$\int d\Omega Y_{\ell m}(\theta, \phi) Y_{\ell' m'}^*(\theta, \phi) = \delta_{\ell\ell'} \delta_{mm'} \quad (3.9)$$

where $d\Omega = \sin\theta d\theta d\phi$.

3.0.4.3 Combined orthonormality of SFB basis

Multiplying ?? by ??, the full SFB basis is orthonormal:

$$\int_0^\infty dr r^2 \int d\Omega j_\ell(kr) Y_{\ell m}(\theta, \phi) j_{\ell'}(k'r) Y_{\ell' m'}^*(\theta, \phi) = \frac{\pi}{2k^2} \delta(k - k') \delta_{\ell\ell'} \delta_{mm'} \quad (3.10)$$

The normalization factors $\sqrt{2/\pi}$ and k in the expansion ?? ensure unit norm.

3.0.4.4 Completeness

The SFB functions form a complete set on the domain $r \in [0, \infty), \theta \in [0, \pi], \phi \in [0, 2\pi)$. Any square-integrable function can be expanded:

$$f(\mathbf{r}) = \sqrt{\frac{2}{\pi}} \int_0^\infty dk \sum_{\ell m} f_{\ell m}(k) k j_\ell(kr) Y_{\ell m}(\theta, \phi) \quad (3.11)$$

with convergence in the L^2 norm (mean-square sense).

Remark 3.1. These orthonormality relations are the foundation for all SFB computations. Any inner product of two fields can be reduced to a simple product of their coefficients, eliminating integrals over $\mathbf{r}$.

See ?? for how to compute SFB coefficients and ?? for power spectrum applications.

Chapter 4

3D Power Spectrum via SFB

Power Spectrum SFB File

This file should contain the content from *power_spectrum_sfb_comprehensive.tex*, refactored with proper LaTeX

The existing comprehensive file covers: - Field decomposition in SFB basis - Power spectrum under SIH - Breaking homogeneity with selection functions - Evolution and bias - Window functions - Gaussian selection functions

****Status****: Ready to be populated from ‘*power_spectrum_sfb_comprehensive.tex*’

****To do****: 1. Copy content from the comprehensive file 2. Wrap in subsection structure 3. Add cross-references to other sections 4. Update all equation labels to follow wiki convention (eq:ps_sfb_{*})

Chapter 5

Weak Lensing: 3D Full-Sky Analysis

5.0.1 Lensing Geometry and the Potential

This section develops the geometry of weak gravitational lensing and introduces the lensing potential.

5.0.1.1 Light bending by gravity

Weak gravitational lensing is the small-angle bending of light rays by intervening masses. In the Born approximation, the bending angle is:

$$\alpha(\theta) = -2 \int dz \frac{dA(z)}{dA(z_s)} \frac{1}{2} \nabla_{\perp} \Phi(z) \quad (5.1)$$

where Φ is the gravitational potential and $\nabla_{\perp}$ is the transverse Laplacian on the sky.

5.0.1.2 The lensing potential

For mathematical convenience, we define the lensing potential ψ that encodes the cumulative bending:

$$\psi(\theta) = 2 \int_0^{z_s} dz \frac{D_A(z) D_A(z_s - z)}{D_A(z_s)} \Phi(z) \quad (5.2)$$

where D_A is angular diameter distance and z_s is source redshift.

The convergence and shear are derived from ψ via derivatives (see ??).

Remark 5.1. The prefactor $D_A(z) D_A(z_s - z) / D_A(z_s)$ is the lensing efficiency: structures at intermediate distances bend rays most effectively, while structures near observer or source have reduced impact.

See ?? for angular diameter distance definitions.

5.0.2 Convergence and Shear: Scalar and Tensor Fields

This section defines the weak lensing fields and their physical interpretation.

5.0.2.1 Convergence: isotropic magnification

The convergence κ is the isotropic magnification/demagnification caused by lensing:

$$\kappa = \frac{1}{2} \nabla_{\perp}^2 \psi \quad (5.3)$$

where $\nabla_{\perp}^2$ is the 2D Laplacian on the sphere.

$\kappa > 0$ means magnification (objects appear brighter), $\kappa < 0$ means demagnification.

****Physical meaning****: κ measures the integrated projected density along the line of sight.

5.0.2.2 Shear: tidal distortion

The shear γ is the anisotropic tidal distortion of source images:

$$\gamma_1 + i\gamma_2 = \left(\frac{\partial^2}{\partial \theta^2} - \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right) \psi - i \frac{2}{\sin \theta} \frac{\partial^2}{\partial \theta \partial \phi} \psi \quad (5.4)$$

This is a spin- ± 2 tensor field (shear stretches images in two perpendicular directions).

5.0.2.3 E-modes and B-modes

Shear decomposes into: - ****E-modes****: Curl-free (gradient-like) component $\rightarrow$ pure from gravitational lensing - ****B-modes****: Divergence-free (curl-like) component $\rightarrow$ zero for pure lensing (systematics diagnostic)

See ?? for harmonic decomposition.

5.0.3 Spin-Weight Spherical Harmonics

This section introduces spin-weighted harmonics needed to decompose tensor fields like shear.

5.0.3.1 Spin weight concept

A quantity with spin weight s transforms as $e^{is\alpha}$ under a rotation by angle α about the radial axis.

- Scalars (convergence): $s = 0 \rightarrow$ use standard $Y_{\ell m}$ - Shear (tidal distortion): $s = \pm 2 \rightarrow$ use $\pm 2 Y_{\ell m}$

5.0.3.2 E/B decomposition

Shear expands as:

$$\gamma = \sum_{\ell m} [\gamma_{\ell m-2}^E Y_{\ell m} + \gamma_{\ell m+2}^B Y_{\ell m}] \quad (5.5)$$

****E-modes**** (γ^E): Curl-free; sourced by lensing potential ****B-modes**** (γ^B): Curl-like; zero for pure gravity (systematics indicator)

5.0.3.3 Harmonics from lensing potential

In harmonic space:

$$\kappa_{\ell m} = \frac{\ell(\ell+1)}{2} \psi_{\ell m} \quad (5.6)$$

$$\gamma_{\ell m}^E = \frac{(\ell-1)(\ell+2)}{2} \psi_{\ell m} \quad (5.7)$$

These algebraic relations follow from Laplacian ($\rightarrow \ell(\ell+1)$) and higher derivatives on the sphere.

See ?? for definitions of κ and γ .

5.0.4 3D Lensing Power Spectrum

This section develops the 3D power spectrum of weak lensing observables in the SFB framework.

5.0.4.1 SFB decomposition of convergence

When sources have extended redshift distribution, decompose in SFB basis:

$$\kappa(\mathbf{r}) = \sqrt{\frac{2}{\pi}} \int_0^\infty dk \sum_{\ell m} \kappa_{\ell m}(k) k j_\ell(kr) Y_{\ell m}(\theta, \phi) \quad (5.8)$$

where r is comoving distance.

5.0.4.2 3D power spectrum definition

$$\langle \kappa_{\ell m}(k) \kappa_{\ell' m'}^*(k') \rangle = C_\ell^\kappa(k, k') \delta_{\ell \ell'} \delta_{m m'} \quad (5.9)$$

$C_\ell^\kappa(k, k')$ is a 2D matrix in wavenumber space at each ℓ .

Under isotropy, $C_\ell(k, k') = P(k) \delta(k - k')$ (single function).

5.0.4.3 Window function approach

With finite survey selection $\phi(r)$:

$$C_\ell^{\text{obs}}(k_1, k_2) = \left(\frac{2}{\pi} \right)^2 \int_0^\infty dk k^2 P(k) W_\ell(k_1, k) W_\ell(k_2, k) \quad (5.10)$$

where the window function encodes both the selection and the lensing kernel.

Remark 5.2. This window function formalism converts complex 3D integrals into 1D convolutions, making numerical evaluation tractable.

See ?? for how these 3D spectra project to observable angular power spectra.

Chapter 6

3D Observables & Applications

6.0.1 Angular Power Spectrum from 3D Power

This section shows how to compute the observable 2D angular power spectra from 3D power spectra.

6.0.1.1 Projection integral

From a 3D field at fixed redshift z , the angular power spectrum is:

$$C_\ell = \int_0^\infty \frac{dk}{2\pi^2} k^2 C_\ell(k) W_\ell^2(k) \quad (6.1)$$

where $W_\ell(k)$ is a window function encoding the survey selection in that redshift bin.

6.0.1.2 Limber approximation

For large ℓ and smooth selection functions, the Limber approximation simplifies:

$$C_\ell^{\text{Limber}} \approx \int_0^\infty \frac{d\chi}{\chi^2} W(\chi) P\left(k = \frac{\ell + 1/2}{\chi}, \chi\right) \quad (6.2)$$

This replaces the integral over wavenumbers with an evaluation at effective wavenumber $k \approx \ell/\chi$.

****Validity**:** $\ell \gg 1$ and $\Delta\chi \ll \chi$ (smooth selection in redshift).

6.0.1.3 Beyond Limber

For accurate treatment at all ℓ , must compute full convolution:

$$C_\ell = \int_0^\infty \frac{dk}{2\pi^2} k^2 P(k) \left[\int_0^\infty d\chi \phi(\chi) j_\ell(k\chi) \right]^2 \quad (6.3)$$

This captures the full 3D structure, not just the diagonal $k = \ell/\chi$ modes.

See ?? for multi-bin cross-spectra.

6.0.2 Redshift Binning and Photo-z Effects

This section addresses how redshift information enters observables through photometric redshifts and redshift bins.

6.0.2.1 Redshift distribution

For a photometric survey, sources are divided into bins i, j with distribution $\frac{dn_i}{dz}(z)$.

The lensing kernel for bin i is:

$$W^i(\chi) = \int dz \frac{dn_i}{dz}(z) W(\chi(z)) \quad (6.4)$$

where $W(\chi)$ is the lensing efficiency per unit distance.

6.0.2.2 Cross-spectrum between bins

Angular power spectra between different redshift bins are:

$$C_\ell^{ij} = \int_0^\infty \frac{dk}{2\pi^2} k^2 P(k) \left[\int d\chi \phi_i(\chi) j_\ell(k\chi) \right] \left[\int d\chi' \phi_j(\chi') j_\ell(k\chi') \right] \quad (6.5)$$

If bins overlap in redshift, $C_\ell^{ij} \neq 0$ even for $i \neq j$ (correlation from common sources).

6.0.2.3 Photo-z uncertainties

Photometric redshifts have errors $\sigma_z \sim 0.01 - 0.05$ (relative). Effects: - Broadens effective $\phi_i(\chi)$ - Causes bin-to-bin cross-contamination - Biases C_ℓ measurements at level $\sim \sigma_z^2$

Must include photo-z covariance in analysis.

See ?? for more on bin combinations.

6.0.3 Integration Kernels for 3D Observables

This section develops the integration kernels that connect underlying matter power spectra to observable weak lensing and clustering.

6.0.3.1 Generic kernel structure

Observable power spectra take the form:

$$C_\ell = \int_0^\infty \frac{dk}{2\pi^2} k^2 P(k) I_\ell^2(k) \quad (6.6)$$

where $I_\ell(k)$ is an integration kernel that encodes the survey geometry and physics.

6.0.3.2 Convergence kernel

For weak lensing convergence:

$$I_\ell^\kappa(k) = \int_0^\infty d\chi W^\kappa(\chi) j_\ell(k\chi) \quad (6.7)$$

where $W^\kappa(\chi)$ is the convergence lensing efficiency (depends on source redshift distribution and angular diameter distances).

6.0.3.3 Density/clustering kernel

For galaxy clustering:

$$I_\ell^\delta(k) = \int_0^\infty d\chi n(\chi) b(\chi) D_+(\chi) j_\ell(k\chi) \quad (6.8)$$

where $n(\chi)$ is galaxy number density, $b(\chi)$ is bias, and $D_+(\chi)$ is growth factor.

6.0.3.4 Numerical evaluation

For accurate predictions: 1. Compute lensing/clustering weights: $W(\chi)$ or $n(\chi)b(\chi)D_+(\chi)$
2. Evaluate Bessel functions $j_\ell(k\chi)$ using recurrence or `SpecialFunctions.jl` 3. Integrate over χ using adaptive quadrature (`QuadGK.jl`) 4. Convolve with power spectrum

See ?? for integration algorithms.

6.0.4 Cross-Spectra Between Observables

This section discusses cross-correlations between different tracers (convergence $\times$ convergence, convergence $\times$ clustering, etc.).

6.0.4.1 Auto vs cross-spectra

- ****Auto-spectrum**** ($i = j$): $C_\ell^{ii} = C_\ell^{AA \times AA}$ (self-correlation) - ****Cross-spectrum**** ($i \neq j$): C_ℓ^{ij} (correlation between different tracers or bins)

6.0.4.2 Convergence $\times$ convergence

Angular power spectrum of weak lensing convergence:

$$C_\ell^{\kappa\kappa} = \int_0^\infty \frac{dk}{2\pi^2} k^2 P(k) \left[\int d\chi W^\kappa(\chi) j_\ell(k\chi) \right]^2 \quad (6.9)$$

Measured by galaxy shape distortions (cosmic shear).

6.0.4.3 Clustering $\times$ clustering

Angular power spectrum of galaxy positions:

$$C_\ell^{\delta\delta} = \int_0^\infty \frac{dk}{2\pi^2} k^2 P(k) \left[\int d\chi W^\delta(\chi) j_\ell(k\chi) \right]^2 \quad (6.10)$$

where $W^\delta(\chi) = n(\chi)b(\chi)D_+(\chi)$.

Measured by galaxy clustering (photometric or spectroscopic).

6.0.4.4 Convergence $\times$ clustering (cross-signal)

Cross-correlation between lensing convergence and galaxy clustering:

$$C_\ell^{\kappa\delta} = \int_0^\infty \frac{dk}{2\pi^2} k^2 P(k) \left[\int d\chi W^\kappa(\chi) j_\ell(k\chi) \right] \left[\int d\chi W^\delta(\chi) j_\ell(k\chi) \right] \quad (6.11)$$

This cross-signal is powerful for cosmology because: - Independent source of 3D information - Cross-covariance with auto-spectra is typically small - Breaks degeneracies in bias and growth

Remark 6.1. Cross-correlations are often cleaner than individual auto-spectra because many systematics (shot noise, PSF effects) don't correlate between independent measurements.

See ?? for how bins combine in cross-spectra.

Chapter 7

Numerical Integration & Methods

7.0.1 Numerical Integration Schemes

This section reviews the primary numerical integration methods used for computing observables.

7.0.1.1 Quadrature-based integration

For 1D integrals of the form:

$$I = \int_a^b f(x)dx \quad (7.1)$$

approximate as weighted sum of function values:

$$I \approx \sum_{i=1}^n w_i f(x_i) \quad (7.2)$$

where x_i are quadrature nodes and w_i are weights.

7.0.1.2 Common quadrature rules

Gauss-Legendre : Optimal for smooth functions; weights/nodes determined by orthogonal polynomials

Gauss-Laguerre : For integrals $\int_0^\infty e^{-x} f(x)dx$ (natural exponential decay)

Adaptive (QuadGK) : Subdivides domain; refines where error is large

Clenshaw-Curtis : Uses Chebyshev nodes; fast FFT-based evaluation

7.0.1.3 Composite integration

For integrals with multiple peaks or wide dynamic range:

$$I = \sum_k \int_{a_k}^{b_k} f(x)dx \approx \sum_k Q_k[f] \quad (7.3)$$

Apply quadrature to each sub-domain $[a_k, b_k]$, then sum.

7.0.1.4 Cosmological examples

Many cosmological integrals have special structure:

- **Comoving distance:** $\chi(z) = \int_0^z \frac{dz'}{E(z')} \rightarrow$ smooth, monotonic integrand
- **Lensing kernels:** $W_\ell(k) = \int d\chi \phi(\chi) j_\ell(k\chi) \rightarrow$ oscillatory Bessel function
- **Power spectrum convolution:** $\int dk k^2 P(k) W(k) \rightarrow$ power-law decay

Each requires adapted strategy (see next sections).

See ?? for specific quadrature algorithms.

7.0.2 Quadrature Strategies for Cosmological Kernels

This section details optimal quadrature strategies for the oscillatory and smooth integrals appearing in cosmology.

7.0.2.1 Gauss-Legendre for smooth integrals

For smooth integrands over finite intervals $[a, b]$:

$$\int_a^b f(x)dx \approx \sum_{i=1}^n w_i^{\text{GL}} f(x_i^{\text{GL}}) \quad (7.4)$$

****Advantages****: - Exact for polynomials up to degree $2n - 1$ - High order convergence for smooth f - $n = 10$ – 20 usually sufficient

****Julia****: ‘using QuadGK; quadgk(f, a, b)’

7.0.2.2 Adaptive quadrature for oscillatory integrals

For integrals with Bessel functions $\int j_\ell(kx)g(x)dx$:

Use adaptive subdivision with automatic error estimation. QuadGK automatically detects oscillations and refines.

Strategy: 1. Start with coarse quadrature 2. Estimate error using Gauss-Kronrod pair 3. If error too large, subdivide interval and recurse 4. Continue until desired tolerance met

****Julia****: ‘quadgk(f, 0, Inf; rtol=1e-8, atol=1e-12)’

7.0.2.3 Transformed integrals for unbounded domains

For $\int_0^\infty f(x)dx$, use change of variables $x = \tan(u)$ or exponential map:

$$\int_0^\infty f(x)dx = \int_0^{\pi/2} f(\tan u) \sec^2 u du \quad (7.5)$$

The Jacobian $\sec^2 u$ vanishes at endpoints, helping quadrature.

7.0.2.4 Chebyshev nodes for fast evaluation

If many evaluations needed (e.g., grid computation), use Chebyshev interpolation: 1. Evaluate f on Chebyshev nodes $x_i = \cos\left(\frac{(2i-1)\pi}{2n}\right)$ 2. Fit Chebyshev polynomial: $f(x) \approx \sum_k c_k T_k(x)$ 3. Integrate analytically: $\int_{-1}^1 f(x)dx \approx \int_{-1}^1 \sum_k c_k T_k(x)dx$

****Advantage****: $O(n)$ evaluation; reuse for many integrands.

See ?? for specialized treatment of Bessel function integrals.

7.0.3 Performance and Optimization

This section addresses computational performance considerations for large-scale numerical integrations.

7.0.3.1 Vectorization and SIMD

Modern processors execute multiple operations in parallel (SIMD). Exploit this:

- **Outer loop over ℓ values**: Compute all $C_\ell(\ell = 0, 1, 2, \dots)$ simultaneously
 - **Batch kernel evaluations**: Evaluate $j_\ell(kr)$ for many (k, r) pairs at once
 - **Avoid scalar loops**: Use array operations (map, reduce) instead of for-loops
- Julia tools**: ‘@turbo’, ‘@tturbo’ (LoopVectorization.jl), ‘@tullio’ (Tullio.jl)

7.0.3.2 Memory access patterns

- **Sequential access**: Much faster than random access (CPU cache efficiency)
- **Cache-friendly layout**: Store data as SoA (structure of arrays) not AoS (array of structures)
- **Pre-allocation**: Avoid repeated memory allocation inside tight loops

7.0.3.3 Bessel function caching

Spherical Bessel functions are expensive. Cache when possible:

- If evaluating $j_\ell(kr_i)$ for multiple i at fixed ℓ, k : compute once, reuse
- Use recurrence relations: $j_{\ell+1}(x) = \frac{(2\ell+1)j_\ell(x) - xj_{\ell-1}(x)}{x}$
- Pre-tabulate for common ℓ values

7.0.3.4 Convergence vs accuracy trade-off

- **Tight tolerance** ($\text{rtol} = 10^{-12}$): More accurate but $10\times$ slower
- **Loose tolerance** ($\text{rtol} = 10^{-6}$): Fast but risky for power spectrum (convolutions amplify errors)
- **Compromise**: $\text{rtol} = 10^{-9}, \text{atol} = 10^{-12}$ (often sufficient)

Remark 7.1. For cosmological predictions competing with observation (1–5% precision), target 0.1% accuracy in integrals to avoid bias.

See ?? for performance measurement techniques.

7.0.4 Benchmarking and Validation

This section discusses how to verify numerical calculations and compare against other methods.

7.0.4.1 Convergence tests

Vary integration tolerance and check convergence:

$$\left| C_\ell^{(\text{tol}_1)} - C_\ell^{(\text{tol}_2)} \right| < 0.01 \times C_\ell^{(\text{tol}_1)} \quad (7.6)$$

If not satisfied, increase tolerance (reduce error).

7.0.4.2 Limber limit validation

For large ℓ (high multipoles), the Limber approximation should match full 3D calculation:

$$\left| \frac{C_\ell^{(\text{Limber})} - C_\ell^{(3D)}}{C_\ell^{(\text{Limber})}} \right| < 0.01 \quad \text{for } \ell > 100 \quad (7.7)$$

Deviations indicate numerical issues or incorrect kernel implementation.

7.0.4.3 Comparison with literature benchmarks

- BLAST paper provides reference C_ℓ values for specific cosmologies - LSST DESC produces public forecasts (available online) - N-body simulations (e.g., Illustris, EAGLE) provide ground-truth power spectra

7.0.4.4 Unit tests

Example: Test that spherical Bessel orthonormality integral equals expected value

$$\text{Result} = \int_0^\infty r^2 j_\ell(kr) j_\ell(k'r) dr \quad (7.8)$$

Should equal $\frac{\pi}{2k^2} \delta(k - k')$ (or close for numerical k, k').

7.0.4.5 Profiling and timing

Use '@time' or '@benchmark' (BenchmarkTools.jl) to identify bottlenecks:

- Profile entire pipeline (Bessel $\rightarrow$ kernel $\rightarrow$ convolution)
- Typical timings: Bessel evaluation (60%), integration (30%), overhead (10%)
- Goal: < 1 second per C_ℓ calculation on modern hardware

Remark 7.2. Consistent timing and convergence with tighter tolerances builds confidence in results. Always compare against a known-good method before trusting new code.

See ?? for more on numerical strategies and ?? for optimization.